



COMPUTER Network

22CSE62 –M.5.2



Module 5 -Application layer and Security mechanisms

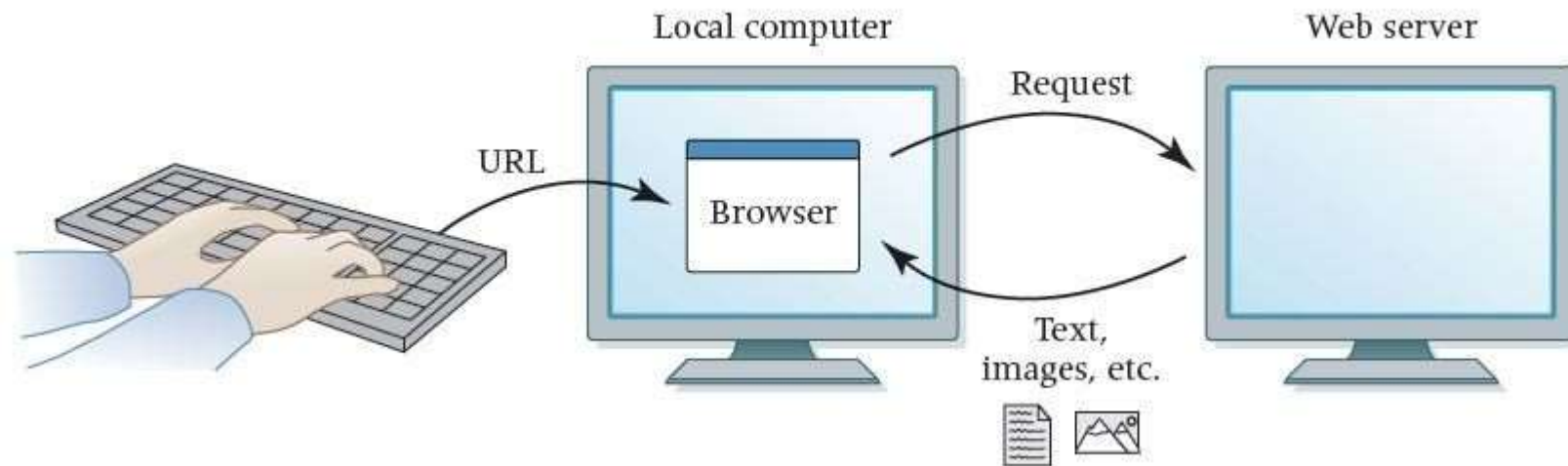


- **Overview of:**
 - Domain Name Space,
 - Remote Logging, Email, FTP, WWW, HTTP
 - Security – Network Layer Security, Application Layer Security



World Wide Web (www)

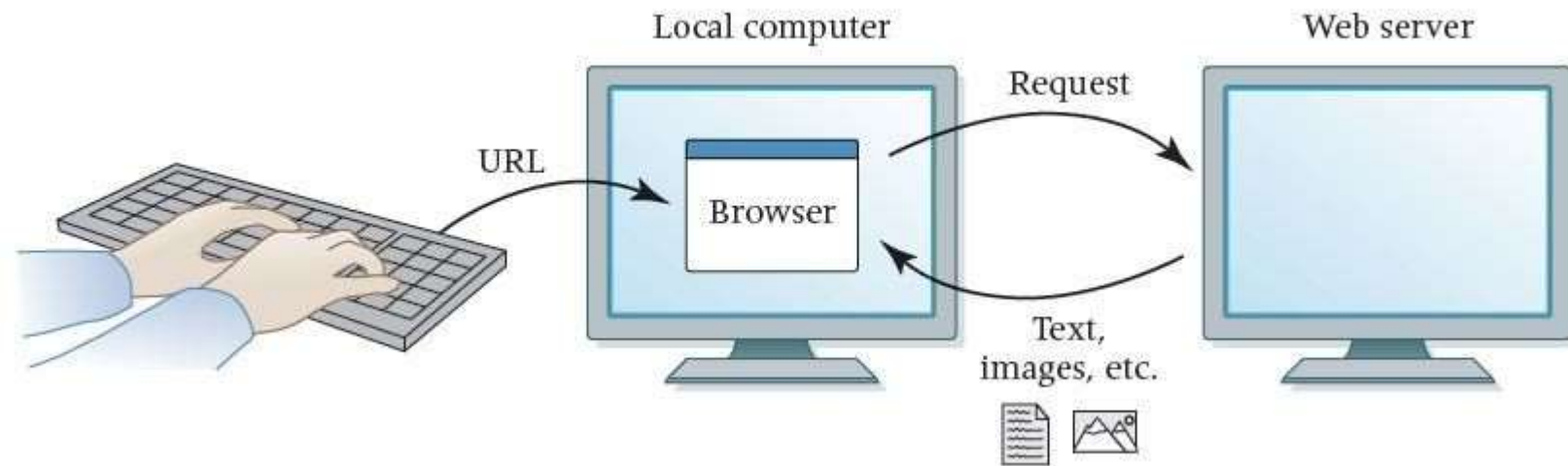
- The World Wide Web (WWW) is a repository of information linked together from points all over the world.
- The WWW has a unique combination of flexibility, portability, and user-friendly features that distinguish it from other services provided by the Internet.





World Wide Web (www)

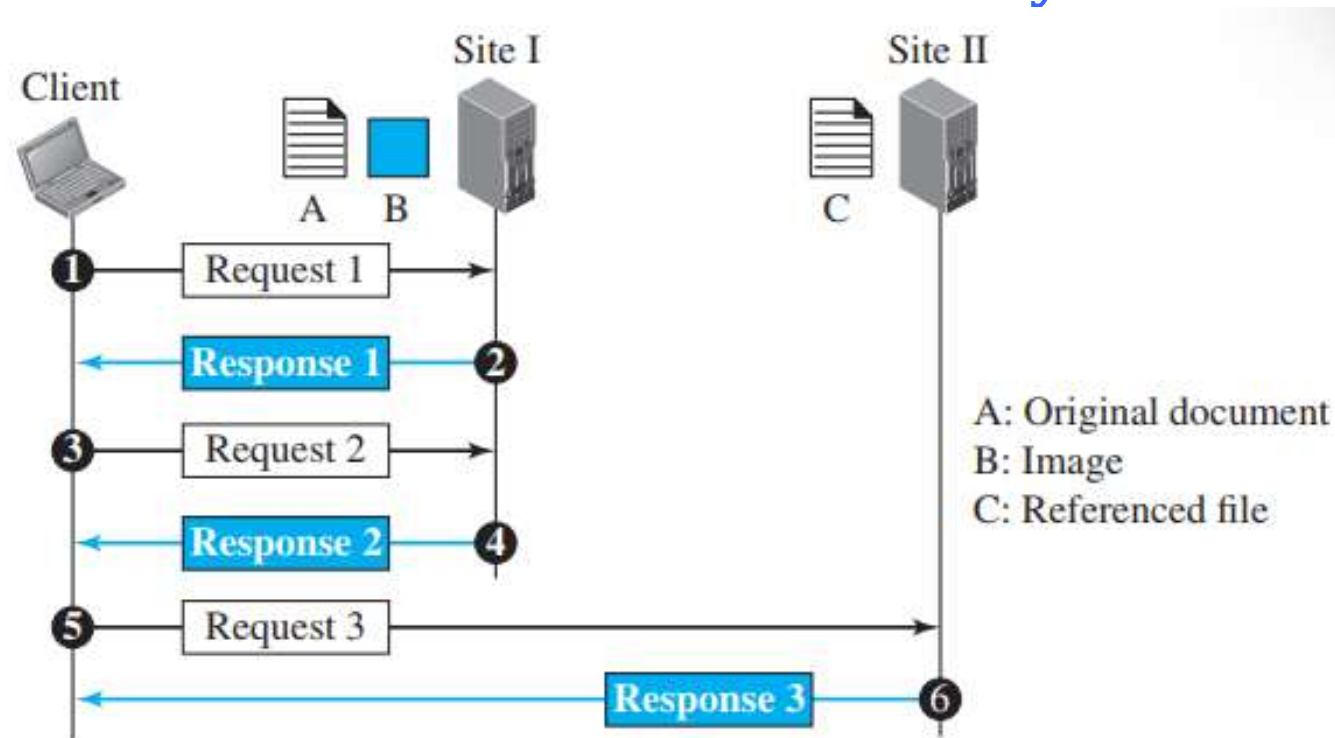
- Website: A collection of related web pages
- Web browser: A software tool that retrieves and displays web pages
- Web server: A computer set up to respond to requests for web pages
- Uniform Resource Locator (URL): A standard way of specifying the location of a Web page, containing the hostname, "/", and a file





Architecture of WWW

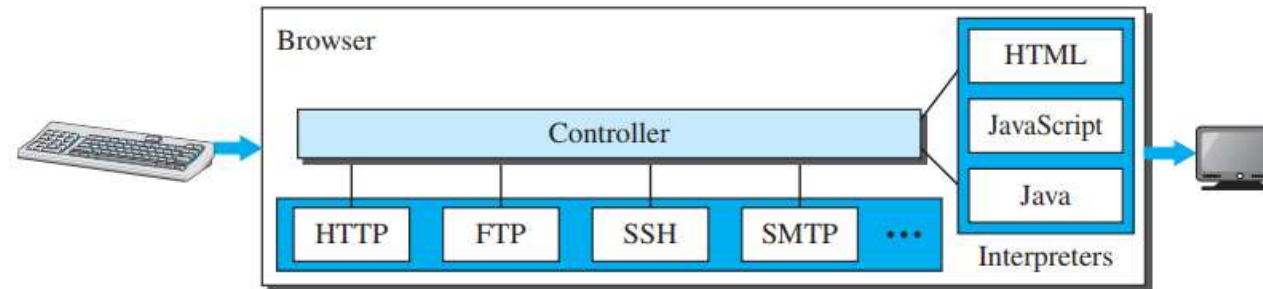
- The WWW today is a distributed client/server service, in which a client using a browser can access a service using a server.
- The service provided is distributed over many locations called sites





Client (Browser)

- Each browser usually consists of three parts:
 - a controller,
 - client protocol, and
 - interpreters
- The controller receives input from the keyboard or the mouse and uses the client protocols to access the document.
- After the document has been accessed, the controller uses one of the interpreters to display the document on the screen.





Server

- The Web page is stored at the server. Each time a client request arrives, the corresponding document is sent to the client.
- To improve efficiency, servers normally store requested files in a cache in memory; memory is faster to access than disk.
- A server can also become more efficient through multithreading or multiprocessing. In this case, a server can answer more than one request at a time.



Uniform Resource Locator

- A URL is defined as any character string that identifies a resource
- The URL defines four things: protocol, host computer, port, and path
- <https://meet.google.com/tsv-qyux-ppi>



protocol://host/path

Used most of the time

protocol://host:port/path

Used when port number is needed



Uniform Resource Locator

- The **protocol** is the client/server program used to retrieve the document.
 - Many different protocols can retrieve a document; among them are FTP or HTTP. The most common today is HTTP.

Common Protocols

- HTTP – HyperText Transfer Protocol
- HTTPS – Secure HTTP
- FTP – File Transfer Protocol



Uniform Resource Locator

- The **host** is the computer on which the information is located.
 - Web pages are usually stored in computers, and computers are given alias names that usually begin with the characters "www". This is not mandatory.

Host

- Host identifies the web server.
- It can be:
 - IP Address
 - Domain Name

Examples

- IP Address: 64.23.56.17
- Domain Name: www.google.com ↗

The host uniquely identifies the computer on the Internet.



Uniform Resource Locator

- The URL can optionally contain the **port number** of the server.
 - If the port is included, it is inserted between the host and the path, and it is separated from the host by a colon.

Port

- Port is a 16-bit integer number.
- It identifies the specific service/application running on the server.

Common Port Numbers

- HTTP → Port 80
- HTTPS → Port 443
- FTP → Port 21

Example

```
http://host:80/
```



Uniform Resource Locator

Path

- Path specifies the location of the file on the server.
- It contains directory names and file name.

Example

```
/top/next/last/myfile
```

Meaning

- myfile is stored inside:
 - last directory
 - under next
 - under top



Types of Web Documents

Web documents are classified into:

- **Static Documents**
- **Dynamic Documents**
- **Active Documents**

Static Documents

- Fixed-content documents stored on a server.
- Same content is sent to every user.
- User cannot modify server content.

Technologies Used

- HTML
- XML
- XHTML
- XSL

Example

- Simple informational web pages



Types of Web Documents

Web documents are classified into

- **Static Documents**
- **Dynamic Documents**
- **Active Documents**

Dynamic Documents

- Created by the server when requested by a browser.
- Content changes dynamically for each request.

Features

- Generated using programs/scripts
- Different users may get different content

Technologies

- JSP (Java Server Pages)
- ASP (Active Server Pages)
- ColdFusion
- CGI

Example

- Current date and time webpage



Types of Web Documents

Web documents are classified into:

- **Static Documents**
- **Dynamic Documents**
- **Active Documents**

Active Documents

- Programs/scripts executed at the client side.
- Used for interaction and animations.

Technologies

- Java Applets
- JavaScript

Examples

- Interactive forms
- Animations
- Games



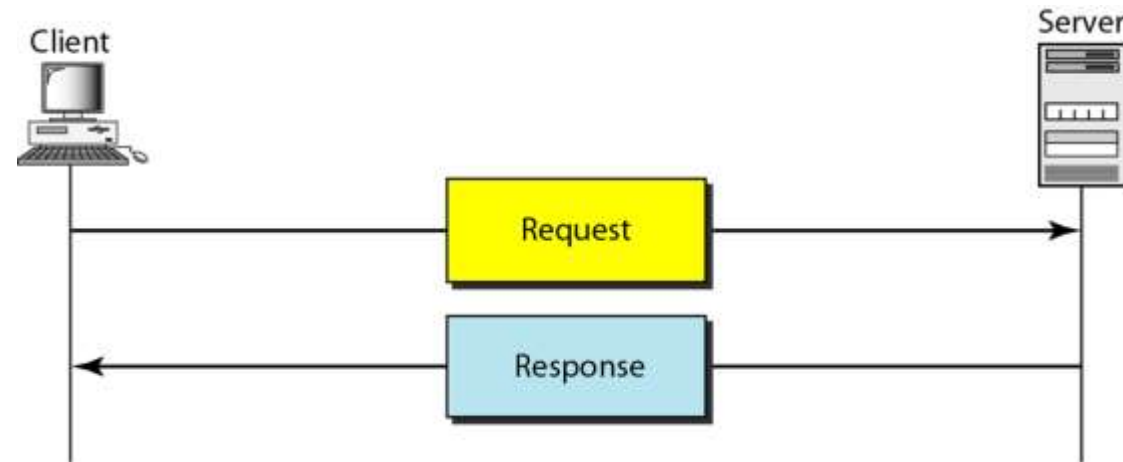
HyperText Transfer Protocol (HTTP)

- The Hypertext Transfer Protocol (HTTP) is a protocol used mainly to access data on the World Wide Web.
- HTTP functions as a combination of FTP and SMTP.
- HTTP uses the services of TCP on well-known port 80.
- It is similar to FTP because it transfers files and uses the services of TCP.
- Unlike FTP there is no separate control connection; only data are transferred between the client and the server



HTTP Transaction

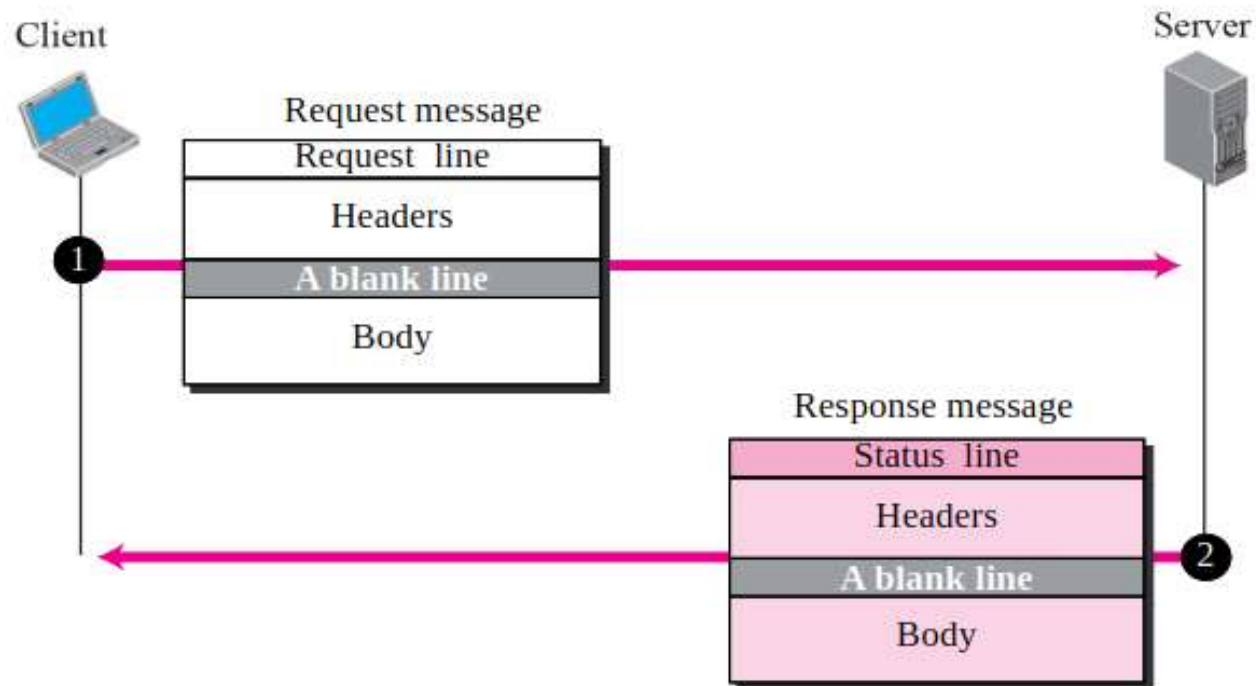
- HTTP uses the services of TCP, however HTTP itself is a stateless protocol.
- The client initializes the transaction by sending a request message.
- The server replies by sending a response message





HyperText Transfer Protocol (HTTP)

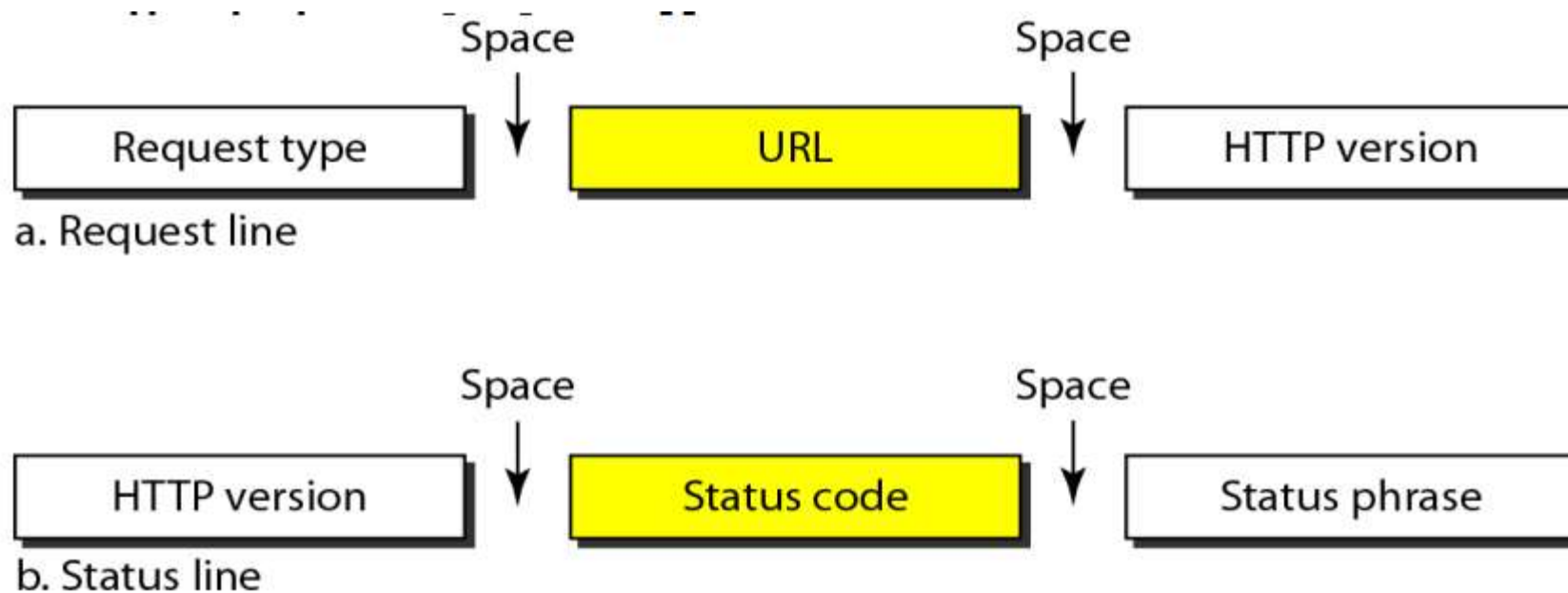
- A request message consists of a request line, a header, and sometimes a body.
- A response message consists of a status line, a header, and sometimes a body





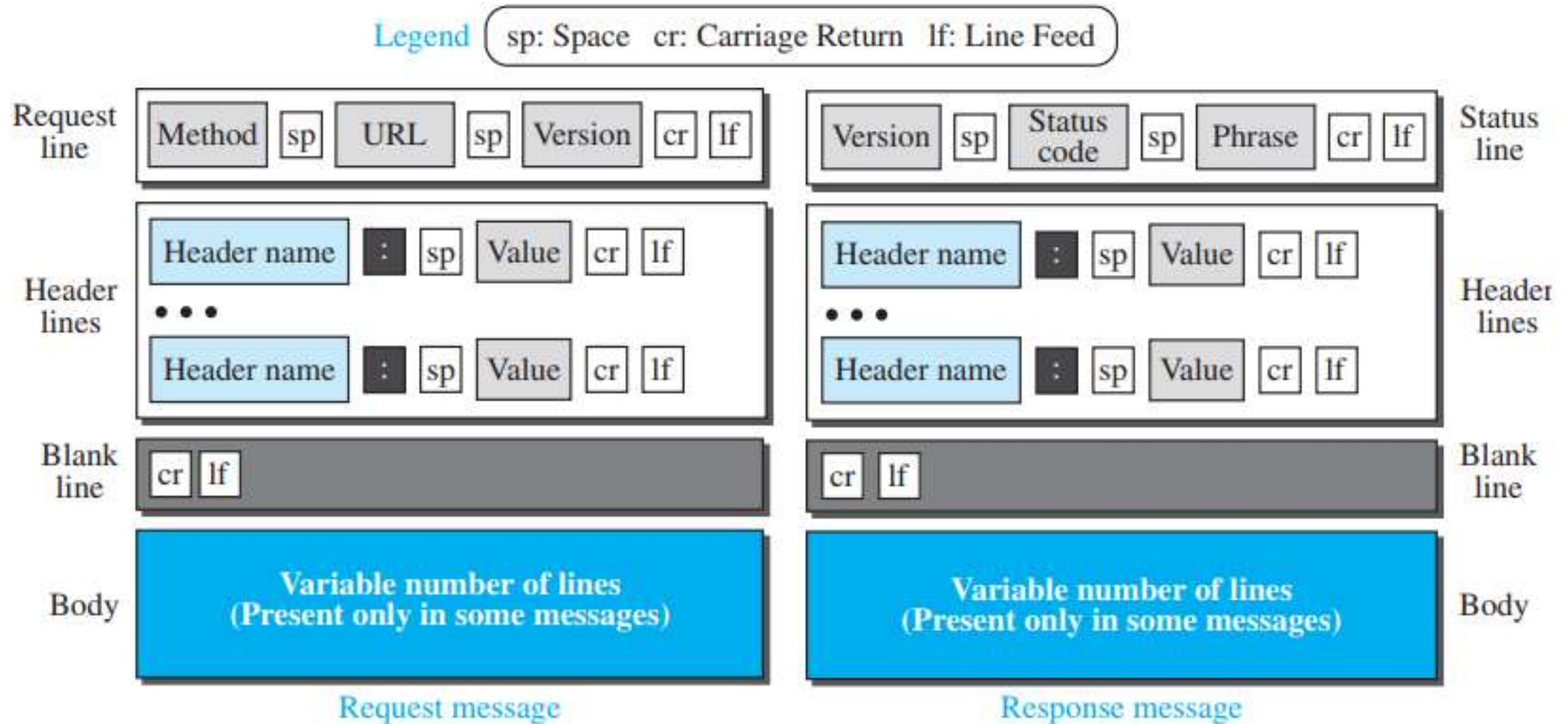
Request and Status Lines:

- The first line in a request message is called a request line.
- The first line in the response message is called the status line.





Format of the request message





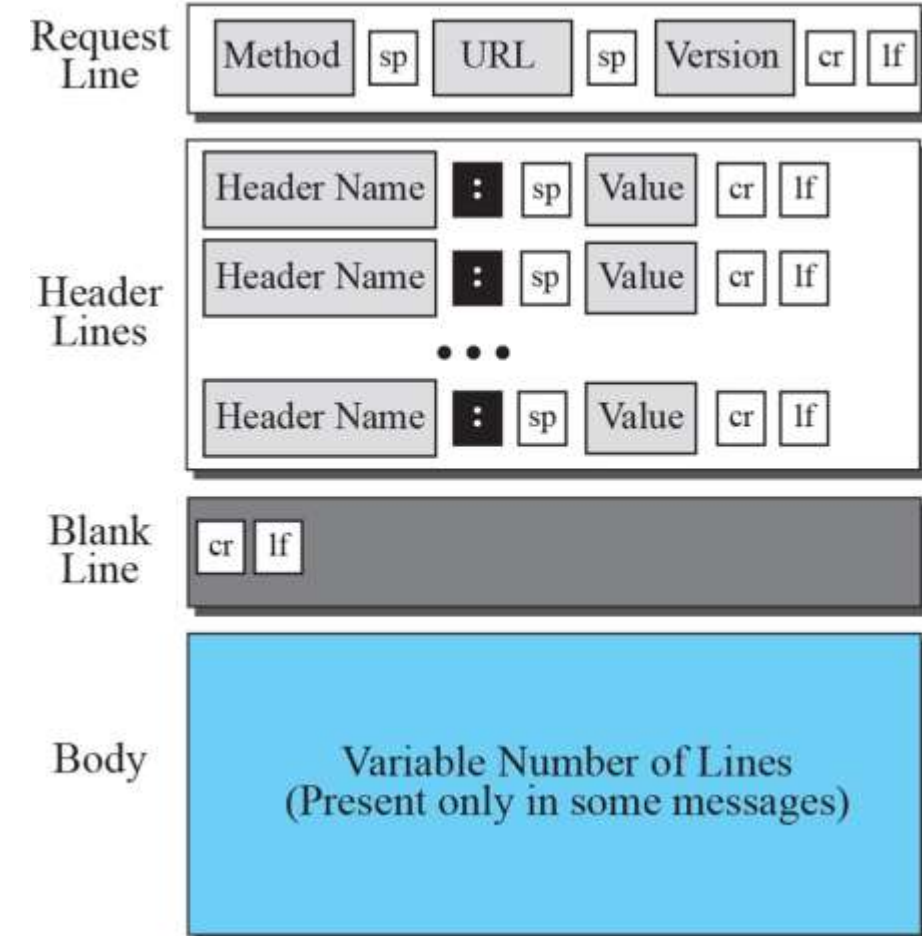
HTTP Message Structure

Both request and response messages contain four sections:

1. First Line
2. Header Lines
3. Blank Line
4. Body

Difference

- Request message → Request Line
- Response message → Status Line





HTTP Request Message Format

Request Message Format

```
Method  URL  Version  
Header-name: Value  
Header-name: Value
```

Body

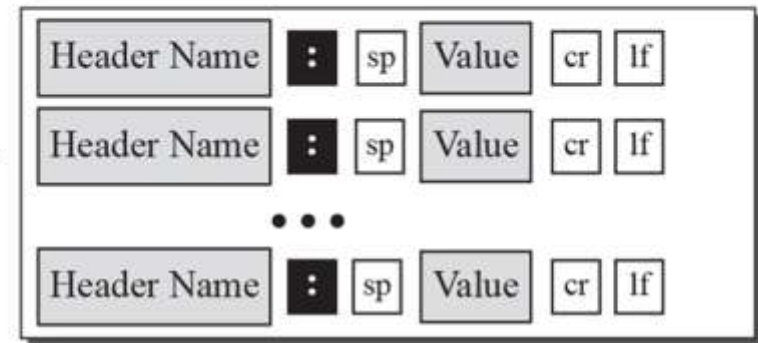
Components

- Method → Type of request
- URL → Resource location
- Version → HTTP version used

Request
Line



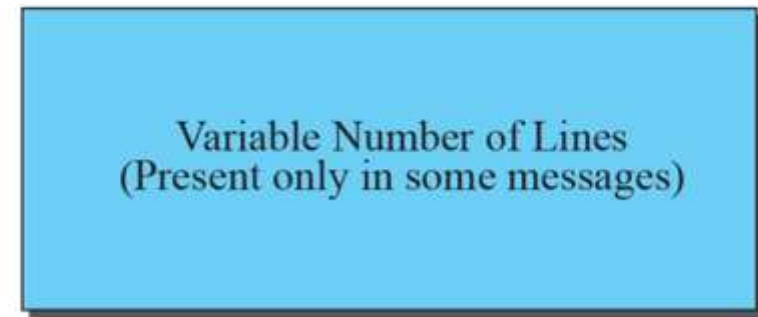
Header
Lines



Blank
Line



Body





Fields in Request Line

1. Method

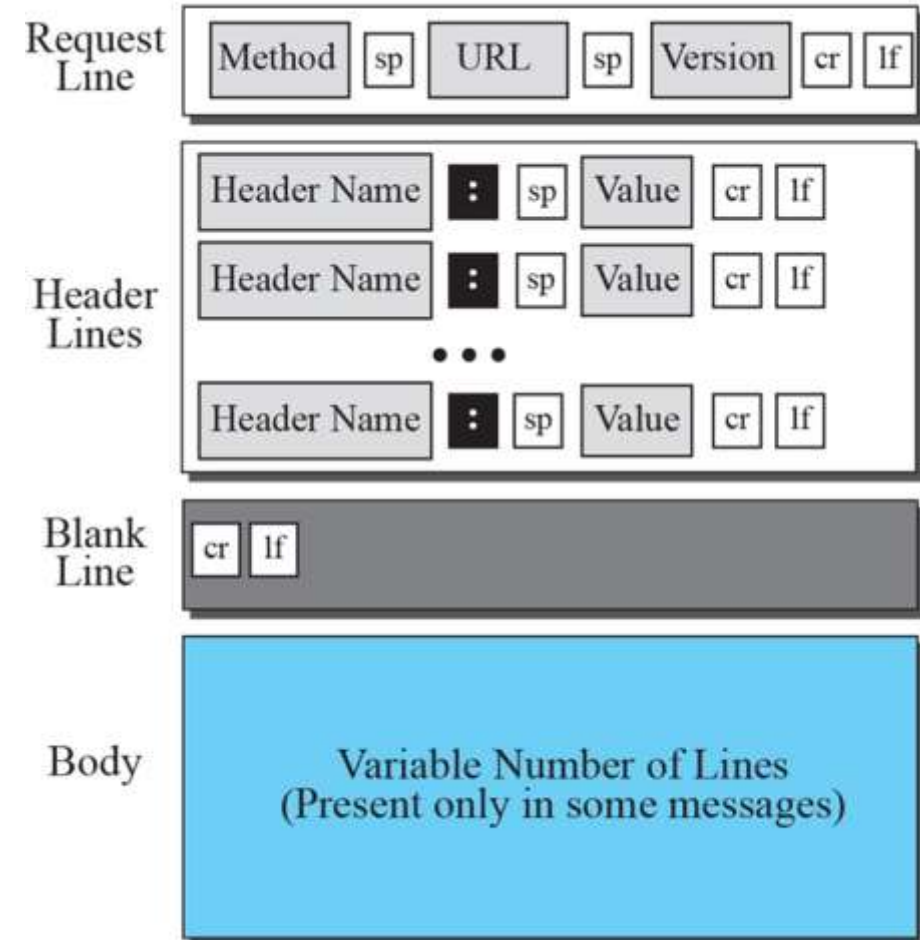
Defines the type of action requested.

2. URL

Specifies address/path of web resource.

3. Version

Indicates HTTP protocol version (e.g., HTTP/1.1)





<i>Method</i>	<i>Action</i>
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
PUT	Sends a document from the client to the server
POST	Sends some information from the client to the server
TRACE	Echoes the incoming request
DELETE	Removes the web page
CONNECT	Reserved
OPTIONS	Inquires about available options



GET and HEAD Methods

GET Method

- Most commonly used
- Retrieves web pages/documents
- Request body usually empty

HEAD Method

- Retrieves only header information
- Used to:
 - Check modification date
 - Validate URL
- Response body is empty

PUT Method

- Uploads new file/web page to server

POST Method

- Sends data to server
- Used for:
 - Form submission
 - Updating web pages
 - Sending comments/data



Request Header Fields

<i>Header</i>	<i>Description</i>
User-agent	Identifies the client program
Accept	Shows the media format the client can accept
Accept-charset	Shows the character set the client can handle
Accept-encoding	Shows the encoding scheme the client can handle
Accept-language	Shows the language the client can accept
Authorization	Shows what permissions the client has
Host	Shows the host and port number of the client
Date	Shows the current date
Upgrade	Specifies the preferred communication protocol
Cookie	Returns the cookie to the server (explained later)
If-Modified-Since	If the file is modified since a specific date

Request Body

- Optional section
- Present mainly in:
 - PUT requests
 - POST requests

Contains

- Uploaded files
- Form data
- Comments
- Web page content



HTTP Response Message Format

Response Message Format

```
Version Status-Code Phrase  
Header-name: Value  
Header-name: Value
```

Body

Components

- Version
- Status Code
- Status Phrase
- Headers
- Body

Status
Line

Version sp Status code sp Phrase cr lf

Header
Lines

Header Name : sp Value cr lf

Header Name : sp Value cr lf

...

Header Name : sp Value cr lf

Blank
Line

cr lf

Body

Variable Number of Lines
(Present only in some messages)

**Table 22.3** Status Codes and Status Phrases

Status Code	Status Phrase	Description
Informational		
100	Continue	The initial part of the request received, continue.
101	Switching	The server is complying to switch protocols.
Success		
200	OK	The request is successful.
201	Created	A new URL is created.
202	Accepted	The request is accepted, but it is not immediately acted upon.
204	No content	There is no content in the body.
Redirection		
301	Moved permanently	The requested URL is no longer used by the server.
302	Moved temporarily	The requested URL has moved temporarily.
304	Not modified	The document has not modified.
Client Error		
400	Bad request	There is a syntax error in the request.
401	Unauthorized	The request lacks proper authorization.
403	Forbidden	Service is denied.
404	Not found	The document is not found.
405	Method not allowed	The method is not supported in this URL.
406	Not acceptable	The format requested is not acceptable.
Server Error		
500	Internal server error	There is an error, such as a crash, at the server site.
501	Not implemented	The action requested cannot be performed.
503	Service unavailable	The service is temporarily unavailable.

Table 22.4 Response Header Names

Header	Description
Date	Shows the current date
Upgrade	Specifies the preferred communication protocol
Server	Gives information about the server
Set-Cookie	The server asks the client to save a cookie
Content-Encoding	Specifies the encoding scheme
Content-Language	Specifies the language
Content-Length	Shows the length of the document
Content-Type	Specifies the media type
Location	To ask the client to send the request to another site
Accept-Ranges	The server will accept the requested byte-ranges
Last-modified	Gives the date and time of the last change



Response Body

- Contains actual document/data
- Examples:
 - HTML page
 - Image
 - Video
 - JSON/XML data

Note

- Body may be absent in error responses or HEAD requests



Example of GET Request

Request

```
GET /usr/bin/image1 HTTP/1.1
Accept: image/gif
Accept: image/jpeg
```

Purpose

- Retrieves image from server
- Client accepts GIF and JPEG formats

Example Response

```
HTTP/1.1 200 OK
Date: Mon, 12 May 2026
Server: Apache
Content-Length: 4520
Content-Type: image/jpeg
```

Response Includes

- Status line
- Header information
- Requested image in body

